SERII NUMERICE

- $\sum u_n$ ABS CONV $\Rightarrow \sum_{n=0}^{\infty} |u_n|$ CONVERGENTĂ

- SERIE GEOMETRICĂ: $\sum q^n$

  CONVERGENTĂ $|q| < 1$ $\quad S = \frac{1}{1-q}$

  DIVERGENTĂ: $|q| \geq 1$

- SERIE ARMONICĂ: $\sum \frac{1}{n^\alpha}$

  CONVERGENTĂ: $\alpha > 1$

  DIVERGENTĂ: $\alpha \leq 1$

CRITERII

1. **NECESAR:**

   $\sum u_n$ CONV $\Rightarrow \lim_{n\to\infty} u_n = 0$

   $\lim_{n\to\infty} u_n \neq 0 \Rightarrow$ DIV

2. **COMPARAȚIE:**

   INEG: $u_n \leq v_n$

   i) $\sum v_n$ CONV $\Rightarrow \sum u_n$ CONV

   ii) $\sum u_n$ DIV $\Rightarrow \sum v_n$ DIV

   LIMITĂ: $\exists \lim \frac{u_n}{v_n} = l$

   $l = 0$ $\sum v_n$ CONV $\Rightarrow \sum u_n$ CONV

   $l \to +\infty$ $\sum v_n$ DIV $\Rightarrow \sum u_n$ DIV

3. RAABE:

$$\lim_{n\to\infty} \sqrt[n]{u_n} = L$$

a) $L < 1 \Rightarrow$ CONV

b) $L > 1 \Rightarrow$ DIV

4. RAABE:

$$\lim_{n\to\infty} \frac{u_{n+1}}{u_n} = L$$

a) $L < 1 \rightarrow$ CONV

b) $L > 1 \Rightarrow$ DIV.

5. INTEGRAL:

$f : [k, \infty) \to [0, \infty)$ — Def pozitiva → $f \downarrow$ → condiție

$k \in \mathbb{N}$

$\rightarrow \int f$ integrabil

$a_n = \int_k^n f(x)\,dx$ — DIV/CONV

$a_n = \int_k^n f(x)\,dx$ — CONV / DIV

Eg (n) CONV $\Leftrightarrow$ a_n CONV.

6. LEIBNIZ:

$\sum_{n=0}^{\infty} (-1)^n u_n$

$u_n \downarrow$ $\xrightarrow{n\to\infty} 0$ $\Big| \Rightarrow$ CONV.

7. DUHAMEL:

↳ cond RAP = 1

$$\lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = L$$

a) $L < 1$ DIV

b) $L > 1$ CONV

8. Logaritmic

$$\lim_{n\to\infty} \frac{\ln \frac{1}{u_n}}{\ln n} = l$$

a) $l < 1 \Rightarrow$ Div

b) $l > 1 \Rightarrow$ Conv

Șiruri + Serii funcții

I Șiruri:

a) CS:

b) CU:

$$\lim_{n\to\infty} f_n(x) = f(x)$$

$$\lim_{n\to\infty} \left(\sup_{x\in E} |f_n(x) - f(x)| \right) = 0$$

(Studiu fct)

Derivez → tabel

$\sup g(x) = g(x^*)$ ↑

Val maximă interval

II Serii:

a) CS:

→ criteriu de la baza de tot.

b) UC:

$|f_n(x)| \leq u_n$ Conv → Sf Conv.

$|f_n(x)| \geq v_n$ Div → Sf Div.

Div

SERIE DE PUTERI:

$$\sum_{n\geq 0} a_n(x-x_0)^n = a_0 + a_1(x-x_0) + \dots$$

RAZA DE CONV

① $\left|\frac{a_{n+1}}{a_n}\right| \to L \qquad R = \frac{1}{L}$

② $\sqrt[n]{|a_n|} \to L \qquad R = \frac{1}{L}$

- $\frac{1}{1-x} = \sum_{n\geq 0} x^n \qquad |x|<1$
- $e^x = \sum_{n\geq 0} \frac{x^n}{n!} \qquad x\in\mathbb{R}$
- $\cos x = \sum_{n\geq 0} (-1)^n \cdot \frac{x^{2n}}{(2n)!} \qquad x\in\mathbb{R}$
- $\sin x = \sum_{n\geq 0} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \qquad x\in\mathbb{R}$
- $\operatorname{ch} x = \sum_{n\geq 0} \frac{x^{2n}}{(2n)!} \qquad x\in\mathbb{R}$
- $\operatorname{sh} x = \sum_{n\geq 0} \frac{x^{2n+1}}{(2n+1)!} \qquad x\in\mathbb{R}$
- $(1+x)^\alpha = 1 + \sum_{n\geq 1} \frac{\alpha(\alpha-1)(\alpha-2)\cdots(\alpha-(n-1))}{n!}\cdot x^n \qquad |x|<1$
- $\ln(1+x) = \sum_{n\geq 1} (-1)^{n+1}\cdot\frac{x^n}{n} \qquad |x|<1$
- $\ln\frac{1}{1-x} = \sum \frac{x^n}{n} \qquad |x|<1$
- $\ln\left(\frac{1+x}{1-x}\right) = 2\sum \frac{x^{2n+1}}{2n+1} \qquad |x|<1$

SUMA SERIEI: $f : [\ ,\ ] \to$

MUST BE CONV

$f(x) = \Sigma$ ——— Serie data

Domeniu

Imagini

!!! Serii alternante !!

ca sa aiungem la o suma exacta

- $\int_0^1 \frac{\operatorname{ch} x - 1}{x^2} dx$ cu 4 zec exacte

$$\operatorname{ch} x = \sum_{n \geq 0} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \dots$$

$$\operatorname{ch} x - 1 = \sum_{n \geq 1} \frac{x^{2n}}{(2n)!}$$

$$\int_0^1 \frac{\operatorname{ch} x - 1}{x^2} = \sum_{n \geq 1} \frac{x^{2n-2}}{(2n)!} = \sum_{n \geq 1} \frac{x^{2n-1}}{(2n)!(2n-1)} \Big|_0^1$$

$$= \sum_{n \geq 1} \frac{1}{(2n)!(2n-1)}$$

$a_n = \frac{1}{(2n \cdot \ )!(2n-1)}$ (toate termenii sunt pozitivi)

Notam cu $S = \int_0^1 \frac{\operatorname{ch} x - 1}{x^2}$

Fie S_n suma partiala

$|S - S_n| < a_{n+1}$

$a_{n+1} < \frac{1}{10^5}$

$n = 0$

$a_0 = \frac{1}{0! \cdot 1}$

$n=1$

$a_1 = \frac{1}{72}$

$n=2$

$a_2 = \frac{1}{3600}$

$n=3$

$a_3 = \frac{1}{35000}$

$n=4 \Rightarrow a_4 = \frac{1}{2822400}$

$n=5 \quad a_5 = \frac{1}{32659200} < 10^{-6}$

$M=4$

$|S - S_4| < a_5$

$-a_5 + S_4 < S < a_5 + S_4$

$S_4 = a_0 + a_1 + a_2 - a_3 + a_4$

SERIA TAYLOR

$$T_n(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

$$\dots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

$e^{-0.2}$ cu 3 zec exacta:

I Fct: $f(x) = e^{-x}$

II Dezv Serie: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$

III Calc. deriv de ord n: $f(x) = e^{-x}$

$$f^{(n)}(x) = (-1)^n e^{-x}$$

IV Se Inlocuieste In Form. TAYLOR + Rest LAGRANGE:

$$f(0,2) = e^{-0,2} = \sum_{k=0}^{n} \frac{(-1)^k}{k!}(0,2)^k + R_n(x)$$

V Rest: $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} =$

$$= \frac{(-1)^{n+1} e^{-c}}{(n+1)!}(0,2)^{n+1}$$

VI $|R_n(x)| < \frac{1}{10^4}$

VII $0 \le c \le 0,2$

$0 \ge -c \ge -0,2 \quad | \cdot e^{\wedge}$

$e^0 \ge e^{-c} \ge e^{-0,2} \quad | \cdot \frac{(0,2)^{n+1}}{(n+1)!}$

$$\frac{e^{-0,2}}{(n+1)!}\left(\frac{1}{5}\right)^{n+1} < |R_n(x)| < \left(\frac{1}{5}\right)^{n+1} \cdot \frac{1}{(n+1)!}$$

VIII

$$n=0 \Rightarrow |R(x)| < \frac{1}{5} \cdot 1 < \frac{1}{10^4} \quad (1)$$

$$n=3 \quad |R(x)| < \frac{1}{15000} < \frac{1}{10^4} \quad (2)$$

IX De calc

$$e^{-0,2} = \sum_{k=0}^{3} \frac{(-1)^k}{(k)!}(0,2)^k \approx 0,8187$$

Aproximam cu 4 zec

$$\int_0^1 \frac{ch x - 1}{x^2} dx$$

$$ch x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \dots$$

$$f(x) = \frac{ch x - 1}{x^2} = \frac{\frac{x^2}{2!} + \frac{x^4}{4!} + \dots}{x^2} =$$

$$= \frac{1}{2!} + \frac{x^2}{4!} + \dots$$

$$I = \int_0^1 f(x)dx = \int_0^1 \left(\frac{1}{2} + \frac{x^2}{24} + \frac{x^4}{720} + \dots\right) dx \approx$$

$$\int_0^1 \frac{1}{2} dx = \frac{1}{2} x \Big|_0^1 = \frac{1}{2} = 0,5$$

$$\int_0^1 \frac{x^2}{24} = \frac{1}{72}$$

$$= \sum_{n=0}^{\infty} \int_0^{\frac{1}{2}} x^{2n}\,dx = \sum_{n=0}^{\infty} \frac{1}{(2n+1)(2n+2)^n}$$

$$\int_0^{\frac{1}{2}} dx = 0{,}5$$

$$\frac{1}{24} x^3 = \frac{1}{24 \cdot 3}$$

$$\frac{1}{720} x^4 = \frac{1}{5 \cdot 720} = \frac{1}{3600}$$

$$\int_0^{\frac{1}{2}} \frac{1}{40320} x^5 \geq \frac{1}{28.2240} < 10^{-5} \quad (A)$$

$$I = \frac{1}{2} + \frac{1}{24 \cdot 3} + \frac{1}{3600} + \frac{1}{28.2240} \geq 0{,}541$$

$$\int_0^{\frac{1}{2}} e^{x^3/12}$$ cu 2zec exacta $\quad x \in [0,1] \Rightarrow \frac{x^3}{2} \in [0, \frac{1}{2}]$

$$R_n(x) = \int \frac{f^{(n+1)}(c)}{(n+1)!} \cdot x^{n+1}$$

$$f(x) = f(0) + f'(0)x + \dots + f^{(n)}(0)\frac{x^n}{n!} + R_n(x)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

$$e^{x^3/12} = \left(1 + \frac{x^3}{2} + \frac{(x^3/12)^2}{2!} + \dots + \frac{(x^3/12)^n}{n!}\right) + \frac{R_n(x)}{x^3}$$

$$x^3 R_n(x) = e^c \cdot \frac{x^3}{2^{n+1}(n+1)!} \cdot x^{3(n+2)}$$

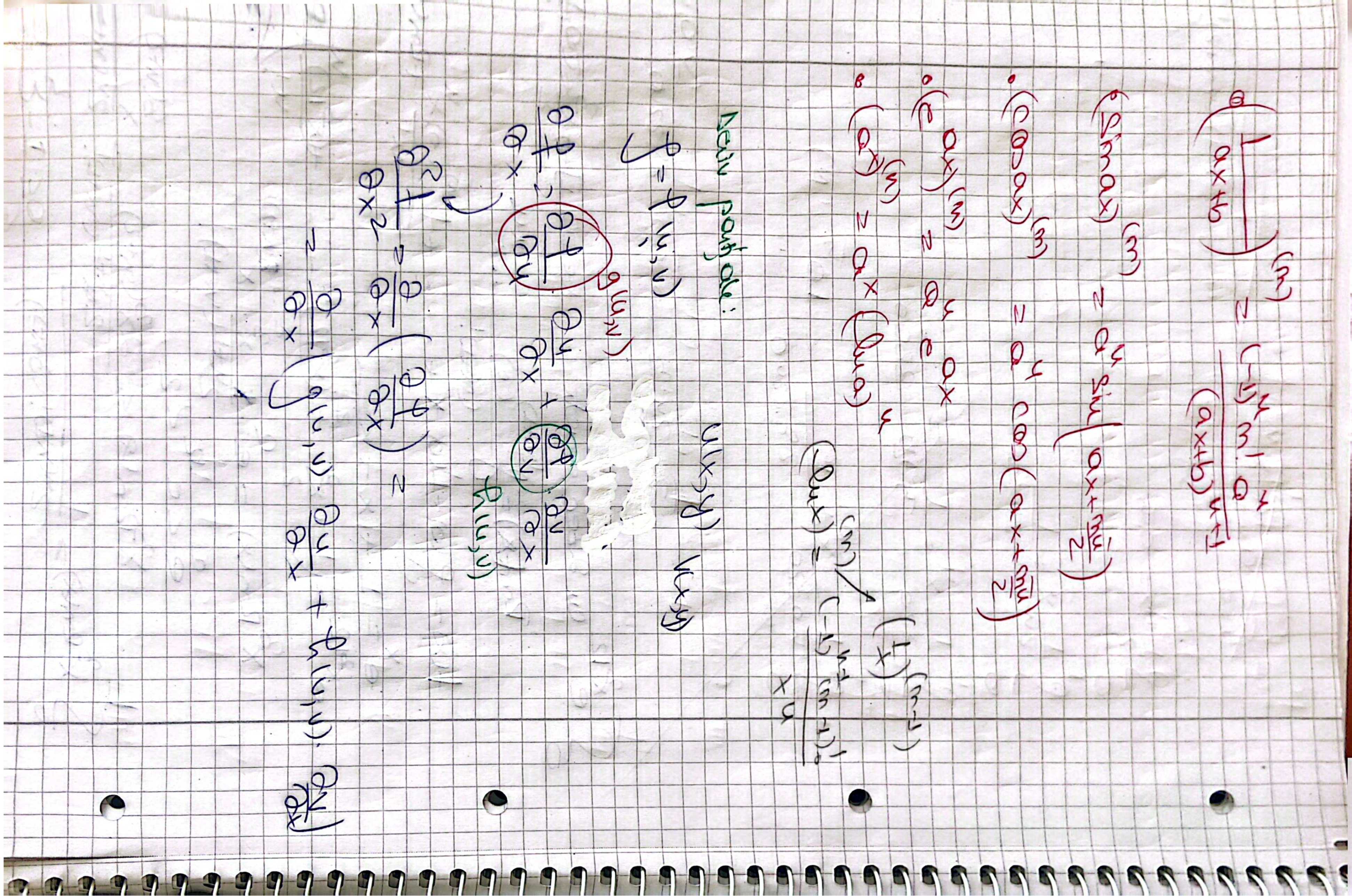
$$e^0 < e^{1/2} < 2$$

$$x^3 R_n(x) \leq \frac{2x^{3(n+2)}}{2^{n+1}(n+1)!} \quad \frac{x^{3(n+2)}}{2^n (n+1)!}$$

$$\int_0^{\frac{1}{2}} x^3 R_n(x) \geq \frac{1}{2^n (n+1)! (3n+7)}$$

$$0 \leq I - \sum_{k=0}^{n} \frac{1}{(3k+1) \cdot k!} \leq \frac{1}{2^{12}}$$

$$\varepsilon \leq$$



INT. SUPERF TRIPLE:

- $\begin{cases} x = \rho\cos t \\ y = \rho\sin t \end{cases}$ $\quad y = \rho \quad t\in[0,2\pi]$ $\quad \rho>0$

$$(x-x_0)^2 + (y-y_0)^2 = R^2$$

- $\begin{cases} x - x_0 = \rho\cos t \\ y - y_0 = \rho\sin t \end{cases}$ $\quad y = \rho \quad t\in[0,2\pi]$ $\quad \rho>0$

$$\frac{(x-x_0)^2}{a^2} + \frac{(y-y_0)^2}{b^2} = 1$$

- $\begin{cases} x - x_0 = a\rho\cos t \\ y - y_0 = b\rho\sin t \end{cases}$ $\quad t\in[0,2\pi]$

$$y = ab\rho \quad \rho\in[0,1]$$

- COORD. CILINDR.:

$$\begin{cases} x = \rho\cos t \\ y = \rho\sin t \\ z = z \end{cases}$$

$$y = \rho \quad \rho>0 \quad t\in[0,2\pi]$$

- COORD. SFERICE:

$$\begin{cases} x = \rho\sin\theta\cos\varphi \\ y = \rho\sin\theta\sin\varphi \\ z = \rho\cos\theta \end{cases}$$

$$\rho>0 \quad \theta\in[0,2\pi] \quad \varphi\in[0,2\pi]$$

$$y = \rho^2\sin\theta$$

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INTEGRALE CURBILINII:

① SEGMENT:

$$\gamma(t): \begin{cases} x(t) = (1-t)x_A + t \cdot x_B \\ y(t) = (1-t)y_A + t \cdot y_B \end{cases} \quad t \in [0,1]$$

② CERC

$$(x-a)^2 + (y-b)^2 = R^2$$

$$\begin{cases} x(t) = a + R\cos t \\ y(t) = b + R\sin t \end{cases} \quad t \in [0, 2\pi]$$

③ ELIPSĂ

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\begin{cases} x(t) = a\cos t \\ y(t) = b\sin t \end{cases} \quad t \in [0, 2\pi]$$

DE PRIMĂ SPEȚĂ

$$\int_\gamma f \, ds = \int_a^b f(\gamma(t)) \cdot \|\gamma'(t)\| \, dt$$

① Masa unui fir Mat.

$$m = \int_\gamma \rho(x,y) \, ds$$

$$\begin{cases} x(t) = f(t) \\ y(t) = g(t) \end{cases}$$

parcurs

$$\sqrt{(\,)^2 + (\,)^2}$$

② centru de greutate

$$R^2 \begin{cases} x_G = \frac{1}{m}\int_A x \cdot \rho(x,y)\,ds \\ y_G = \frac{1}{m}\int_A y \cdot \rho(x,y)\,ds \end{cases}$$

$$R^3 \quad z_G = \frac{1}{m}\int_A z \cdot \rho(x,y,z)\,ds$$

! $\int_\gamma$ in omogen $\Rightarrow \rho(x,y,z) = k\,ct.$

STEIA A DOUA:

R^2: $\alpha = P\,dx + Q\,dy$

$\gamma(t) = (x(t), y(t))$

$$\int_\gamma \alpha = \int_a^b (P(\gamma(t)) \cdot x'(t) + Q(\gamma(t))\,y'(t))\,dt$$

- $\gamma(t) \to$ deprez param $\Rightarrow \int$ ORIENTAT
- $\gamma(t) \to \cap$ a doua supraf $\to \int$ inchisa

$$\int_\gamma \alpha \overset{\text{mat}}{=} \int_\gamma \vec{v} \cdot d\vec{r}$$

↑ circulatia campului

$\vec{v} = \vec{F}$ camp de $\int_\gamma \Rightarrow$ L.M.

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$$\frac{\partial P}{\partial z} = \frac{\partial \rho}{\partial x}$$

$$\frac{\partial P}{\partial y} = \frac{\partial \rho}{\partial z}$$

$$\frac{\partial P}{\partial x} = \frac{\partial \rho}{\partial y}$$

$$\frac{\partial P}{\partial z} = \frac{\partial \rho}{\partial x} \qquad \frac{\partial P}{\partial y} = \frac{\partial \rho}{\partial z} \qquad \frac{\partial P}{\partial x} = \frac{\partial \rho}{\partial y}$$

$$\frac{\partial \rho}{\partial z} = 2 \qquad \frac{\partial \rho}{\partial y} = 0 \qquad \frac{\partial \rho}{\partial x} = P$$

$$\rho = Pdx + Qdy + Rdz$$

$$\rightarrow \text{EXACT} \qquad \rightarrow \text{INEXACT}$$

INTEGRALE DE SUPRAFAȚĂ

$$\sigma(u,v) = (x(u,v), y(u,v), z(u,v))$$

$$\vec{\sigma}(u,v) = x(u,v)\vec{i} + y(u,v)\vec{j} + z(u,v)\cdot\vec{k}$$

→ PARAMETRIZARE CARTEZIANĂ:

$$(x,y,z) \longrightarrow (x,y,g(x,y))$$

$$\begin{cases} u = x \\ v = y \end{cases} \qquad \sigma(x,y) = (x,y,g(x,y))$$

$$\vec{\sigma}(x,y) = x\vec{i} + y\vec{j} + g(x,y)\vec{k}$$

SPETA I:

① PARAM CARTEZIENE:

- $\sigma(x,y,z) = \sigma(x,y,g(x,y))$
- $p = \dfrac{\partial g}{\partial x} \qquad q = \dfrac{\partial g}{\partial y}$
- VECTORI TANGENȚI
  - $\vec{\sigma x} = 1\vec{i} + 0\vec{j} + p\vec{k}$
  - $\vec{\sigma y} = 0\vec{i} + 1\vec{j} + q\vec{k}$
- $N(x,y) = \vec{\sigma x} \times \vec{\sigma y} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & p \\ 0 & 1 & q \end{vmatrix} = -p\vec{i} - q\vec{j} + 1\vec{k}$

$$\boxed{N(-p,-q,1)}$$

$$\rightarrow \iint_S f(x,y,z)\,dA = \iint_D f(x,y,g(x,y))\sqrt{1+p^2+q^2}\,dx\,dy$$

ARIA

② PARAM GENERALĂ:

- $\sigma(u,v) = (x(u,v), y(u,v), z(u,v))$
- $\vec{\sigma u} = \dfrac{\partial \sigma}{\partial u} = \dfrac{\partial x}{\partial u}\vec{i} + \dfrac{\partial y}{\partial u}\vec{j} + \dfrac{\partial z}{\partial u}\vec{k}$
- $\vec{\sigma v} = \dfrac{\partial \sigma}{\partial v} = \dfrac{\partial x}{\partial v}\vec{i} + \dfrac{\partial y}{\partial v}\vec{j} + \dfrac{\partial z}{\partial v}\vec{k}$
- $N(u,v) = \vec{\sigma u} \times \vec{\sigma v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix}$

$$\iint_S \vec{F}\cdot d\vec{S} = \iint_D \vec{F}(\vec{r}(u,v))\cdot(\vec{r}_u \times \vec{r}_v)\,du\,dv$$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} = A\vec{i} - B\vec{j} + C\vec{k}$$

$$A = \begin{vmatrix} \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \quad B = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \quad C = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\text{Ομάδα} = \iint_D \|\vec{N}(u,v)\|\,du\,dv$$

$$\int_S f(x,y,z)\,dS = \iint_D f(x(u,v),y(u,v),z(u,v))\,\|\vec{N}(u,v)\|\,du\,dv$$

• Παράδειγμα Εφαρμογή $\int_S f(x,y,z)\,dS$

• Ορισμένα $\vec{r}(u,v) = (x(u,v), y(u,v), z(u,v))$

Γήεμμ, Λήξεται!

$$\vec{N} = \vec{r}_u \times \vec{r}_v \quad \|\vec{N}\| = \sqrt{A^2+B^2+C^2}\,du\,dv$$

• Παράδειγμα $\|\vec{N}\|$ υπολογισμός

$$\vec{r}(u,v) = (x(u,v), y(u,v), z(u,v))$$

$$\vec{r}_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right)$$

$$\vec{r}_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix}$$

$$\|\vec{N}\| = \sqrt{A^2+B^2+C^2}$$

$$\iint_S f(x,y,z)\,dS = \iint_D f(x(u,v),y(u,v),z(u,v))\,\|\vec{N}\|\,du\,dv$$

$$\iint_S \vec{F}\cdot d\vec{S} = \iint_D \vec{F}(\vec{r}(u,v))\cdot(\vec{r}_u \times \vec{r}_v)\,du\,dv$$

$$\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} = A\vec{i} - B\vec{j} + C\vec{k}$$

$$A = \begin{vmatrix} \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \quad B = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial z}{\partial v} \end{vmatrix} \quad C = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$\vec{N} = \vec{r}_u \times \vec{r}_v \quad \|\vec{N}\| = \sqrt{A^2+B^2+C^2}\,du\,dv

FORMULE INTEGRALE:

- GRADIENT:

$$\text{grad } f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}$$

- DIVERGENTA:

$$\text{div } \vec{v} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

- ROTORUL:

$$\text{rot } \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\vec{k}$$

STOKES:

Int curba spatiu $\rightarrow$ Int. suprafata

$$\alpha = P\,dx + Q\,dy + R\,dz$$

$$\vec{v} = P\vec{i} + Q\vec{j} + R\vec{k}$$

$$\text{Munca} = \int_{\Gamma} \vec{v}\,d\vec{r} = \iint_{S} \text{rot}\,\vec{v} \cdot \vec{n}\,dS$$

GAUSS-OSTROGRADSKI: $\iint \text{suprat } \Sigma \rightarrow \iiint$

$$w = P\,dy\,dz + Q\,dx\,dz + R\,dx\,dy$$

$$\vec{v} = P\vec{i} + Q\vec{j} + R\vec{k}$$

$$\iint_{\Sigma} w = \iint P\,dy\,dz + Q\,dx\,dz + R\,dx\,dy =$$

$$= \iiint \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz$$

$$\iint_{\Sigma} \vec{v} \cdot \vec{n}\,d\sigma = \iiint \text{div}\,\vec{v}\,dx\,dy\,dz$$

INT. IMPROPRII:

CONVERGENTA:

COMPLA LIMITA:

- $[a,+\infty)$ $(a>0)$: $g(x)=\frac{1}{x^\alpha}$
- $(-\infty,b]$ $(b<0)$: $g(x)=\frac{1}{(-x)^\alpha}$

$\left.\begin{array}{l}\alpha>1 \text{ conv}\\ \alpha\leq 1 \text{ div}\end{array}\right\}$

- $[a,b)$: $g(x)=\frac{1}{(b-x)^\alpha}$
- $(a,b]$: $g(x)=\frac{1}{(x-a)^\alpha}$

$\left.\begin{array}{l}\alpha<1 \text{ cond}\\ \alpha\geq 1 \text{ div}\end{array}\right\}$

CU PARAMETRU:

① a) f bine def

b) f continua

c) $\exists\ f'_y$ continua $\frac{\partial f}{\partial y}$

② a) $F(y)=\int_a^b \frac{\partial f}{\partial y}(x,y)\,dx$

b) $F(y)=\int F(y)\,dy+C$

$tg\frac{x}{2}=t$ $\Rightarrow dt=(1+t^2)dx$

$\sin^2 x=\frac{t^2}{1+t^2}$ $\cos^2 x=\frac{1}{1+t^2}$

$dt=\frac{1}{\cos x}dx$

$tg\frac{x}{2}=t$

$\cos x=\frac{1-t^2}{1+t^2}$

$x=2\,arctg\,t$

$dx=\frac{2}{1+t^2}dt$

LEIBNIZ

$\Rightarrow F'(y)=\int_a^b \frac{\partial f}{\partial y}(x,y)\,dx$

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Exercice 1 :

1) $\frac{\partial f}{\partial x} = 0$ et $\frac{\partial f}{\partial y} = 0$

Il faut résoudre ce système pour trouver les points critiques

$\Delta = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix}$

Si $\Delta > 0$ et $\frac{\partial^2 f}{\partial x^2} > 0$ alors f admet un minimum local

Si $\Delta > 0$ et $\frac{\partial^2 f}{\partial x^2} < 0$ alors f admet un maximum local

Si $\Delta < 0$ alors f n'admet ni un maximum ni un minimum local

Si $\Delta = 0$ on ne peut rien conclure

2) $H_f(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}$

$H_f(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{pmatrix}$

Pour $\Delta_1 = \frac{\partial^2 f}{\partial x^2}$, $\Delta_2 = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial x \partial z} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} & \frac{\partial^2 f}{\partial y \partial z} \\ \frac{\partial^2 f}{\partial z \partial x} & \frac{\partial^2 f}{\partial z \partial y} & \frac{\partial^2 f}{\partial z^2} \end{vmatrix}$

Si $\Delta_1 > 0$, $\Delta_2 > 0$, $\Delta_3 > 0$ alors f admet un minimum local

Si $\Delta_1 < 0$, $\Delta_2 > 0$, $\Delta_3 < 0$ alors f admet un maximum local

Si $\Delta_2 < 0$ alors f n'admet ni un maximum ni un minimum local

Si $\Delta_3 = 0$ on ne peut rien conclure

INT. GAMMA:

$$\Gamma(p) = \int_0^{+\infty} e^{-x} x^{p-1} dx$$

$t = ax + b$

1) $\Gamma(1) = 1$
2) $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
3) $\Gamma(p+1) = p \cdot \Gamma(p)$
4) $\Gamma(n) = (n-1)!$ $n \in \mathbb{N}$
5) $\Gamma\left(n+\frac{1}{2}\right) = 1 \cdot 3 \cdot \ldots \cdot (2n-1)\frac{\sqrt{\pi}}{2^n}$
6) $\Gamma(p)\Gamma(1-p) = \frac{\pi}{\sin \pi p}$

INT BETTA:

1) $B(p,q) = \int_0^1 x^{p-1}(1-x)^{q-1} dx$
2) $B(p,q) = \int_0^{+\infty} \frac{x^{p-1}}{(1+x)^{p+q}} dx$
3) $B(p,q) = \int_0^1 \frac{x^{p-1} + x^{q-1}}{(1+x)^{p+q}} dx$

1) $B(p,q) = \frac{\Gamma(p) \cdot \Gamma(q)}{\Gamma(p+q)}$
2) $B(p,q) = B(q,p)$
3) $B(p,q) = \frac{p-1}{p+q-1} B(p-1, q)$

ELIPSOID: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$

PARABOLOID ELIPTIC: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$

HIPERBOLIC: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{z}{c}$

CON: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$

HIPERBOLOID cu O PÂNZĂ: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 =$

— " — 2 — " — : $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{a^2} - 1 =$

CILINDRU ELIPTIC: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$

— " — HIPERBOLIC: $\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0$

— " — PARABOLIC: $y^2 = 2px$

Calculați, și tu:

④ $\int_{\delta} (x^2+5y+2z)\,dx + (y^2-3z+5x)\,dy +$

$+ (z^2+2x-3y)\,dz$

$\delta: \begin{cases} x^2+y^2=4z & z \le 8 \\ x-y=0 \\ x=y \end{cases}$

$2x^2=4z$

$x^2=2z$

$z \in [0,8] \Rightarrow 2z \in [0,16]$

$x^2 \in [0,16]$

$\delta: \begin{cases} x=t \\ y=t \\ z=\frac{t^2}{2} \end{cases}$

$\delta(t): \begin{cases} x'(t)=1 \\ y'(t)=1 \\ z'(t)=t \end{cases}$ $x \in [-4,4]$

$\int_{-4}^{4} \left(t^2+5t+\frac{t^2}{2}\right)\cdot 1 + \left(t^2-\frac{3t^2}{2}+5t\right)\cdot 1 +$

$+\left(\frac{t^4}{4}+2t-3t\right)\cdot t \; dt =$

$= 2\int_{-4}^{4} \left(2t^2+5t-\frac{t^2}{2}+5t+\frac{t^5}{4}+2t^2-3t^2\right) dt =$

$= 2\int_{-4}^{4} \left(t^2-\frac{t^2}{2}+10t+\frac{t^5}{4}\right) dt =$

$= 2\int_{-4}^{4} \left(\frac{t^2}{2}+\frac{t^5}{4}+10t\right) dt =$

$= 10t^2\Big|_{-4}^{4} + \frac{1}{4}\cdot\frac{t^6}{6}\Big|_{-4}^{4} + \frac{t^3}{3\cdot 2}\Big|_{-4}^{4} =$

$= - - - -$

(b) Calculați aria suprafeței

$\Sigma_1 : x^2+y^2+z^2=4$, $z \geq 0$

care se află în int. cilindrului

$x^2+y^2=2y$

$\Sigma: \begin{cases} x=x \\ y=y \\ z=\sqrt{4-x^2-y^2} \end{cases}$

$\frac{\partial z}{\partial x} = -\frac{x}{\sqrt{4-x^2-y^2}}$

$\frac{\partial z}{\partial y} = -\frac{y}{\sqrt{4-x^2-y^2}}$

$$\text{aria}\,\Sigma = \iint_{x^2+y^2\leq 2y} \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2}\,dx\,dy =$$

$$= \iint_{x^2+y^2\leq 2y} \sqrt{1+\frac{x^2}{4-x^2-y^2}+\frac{y^2}{4-x^2-y^2}}\,dx\,dy =$$

$$= \iint_{x^2+y^2\leq 2y} \sqrt{\frac{4-x^2-y^2+x^2+y^2}{4-x^2-y^2}}\,dx\,dy =$$

$$= \iint_{x^2+y^2\leq 2y} \frac{2}{\sqrt{4-x^2-y^2}}\,dx\,dy$$

$\begin{cases} x=\rho\cos t \\ y=\rho\sin t \end{cases}$ $\rho>0$, $t\in[0,2\pi]$

$x^2+y^2\leq 2y$

$\rho \leq 2\sin t$, $\rho>0$ $\Rightarrow t\in$

$$\Rightarrow \text{aria } \mathcal{D} = \int_0^{\pi} \left(\int_0^{2\sin t} \frac{2x}{\sqrt{4-x^2}}\, dx \right) dt.$$

$$\int_0^{2\sin t} \frac{2x}{\sqrt{4-x^2}}\, dx = -2\sqrt{4-x^2}\Big|_0^{2\sin t} =$$

$$u = 4 - x^2 \Rightarrow du = -2x\,dx$$

$$-du = 2x\,dx$$

$$= -2\sqrt{4-4\sin^2 t} - (-2 \cdot 2) =$$

$$= -2\sqrt{4-4\sin^2 t} + 4$$

$$\Rightarrow \int_0^{\pi} 4 - 2\sqrt{4-4\sin^2 t}\, dt =$$

$$= \int_0^{\pi} 4 - 4\sqrt{1-\sin^2 t}\, dt =$$

$$= 4\pi - 4 \cdot \int_0^{\pi} |\cos t|\, dt =$$

$$= 4\pi - 4 \cdot \int_0^{\pi/2} \cos t\, dt + \int_{\pi/2}^{\pi} \cos t\, dt =$$

$$= 4\pi - 4 \cdot \sin t\Big|_0^{\pi/2} + 4\sin t\Big|_{\pi/2}^{\pi} =$$

$$= 4\pi - 4 \cdot 1 + 4 \cdot 0 =$$

$$= 4\pi - 4$$

Lungimea curbei

⑤ $x = e^t \cos 2t$

$y = e^t \sin 2t$

$z = 2e^t$

$$x'(t) = e^t(1 - 2\sin 2t)$$

$$y'(t) = e^t(1 + 2\cos 2t)$$

$$z'(t) = 2e^t$$

$$\sqrt{e^{2t}(1 - 4\sin 2t + 4\sin^2 2t)}$$

$$x'(t) = e^t(\cos 2t - 2\sin 2t)$$

$$y'(t) = e^t(\sin 2t + 2\cos 2t)$$

$$z'(t) = 2e^t$$

$$\sqrt{e^{2t}(\cos^2 2t - 4\cos 2t \cdot \sin 2t + 4\sin^2 2t) + e^{2t}(\sin^2 2t + 4\cos 2t \sin t + 4\cos^2 t) + 4e^{2t}} =$$

$$\Rightarrow \sqrt{e^{2t}[5(\cos^2 2t + \sin^2 2t)] + 4e^{2t}} =$$

$$= \sqrt{9e^{2t}} = 3e^t$$

$$\int_0^1 3e^t dt = 3e - 3$$

(6) 1.5

(1) (2) (3) (4) (5) (6) (7) (8) (9) (10)

$$i) \iint_S x(z^2+3x^2)\,dy\,dz + y(x^2+3y^2)\,dz\,dx + z(y^2+3z^2)\,dx\,dy$$

$$\boxed{x^2+y^2+z^2=2}$$ Fata ext

$$P = x\cdot z^2 + 3x^3$$
$$Q = y\cdot x^2 + 3y^3$$
$$R = z\cdot y^2 + 3z^3$$

$$\operatorname{div}\vec{v} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$= z^2 + 9x^2 + x^2 + 9y^2 + y^2 + 9z^2$$

$$= 10x^2 + 10y^2 + 10z^2$$

$$\Phi_{lat} = \iiint_K 10(x^2+y^2+z^2)\,dx\,dy\,dz$$

$$\begin{cases} x = \vartheta \sin\theta\cos\varphi \\ y = \vartheta \sin\theta\sin\varphi \\ z = \vartheta\cos\theta \end{cases}$$

$$J = \vartheta^2\sin\theta$$

$$\vartheta \in [0, \sqrt{2}] \qquad \theta \in [0, \tfrac{\pi}{2}] \qquad \varphi \in [0, \tfrac{\pi}{2}]$$

$$Flux = 10 \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \left(\int_0^{\sqrt{2}} \vartheta^4 \sin\theta\, d\vartheta \right) d\varphi\, d\theta =$$

$$= \frac{10(\sqrt{2})^5}{5} \int_0^{\frac{\pi}{2}} \left(\int_0^{\frac{\pi}{2}} \sin\theta\, d\varphi \right) d\theta$$

$$= (\sqrt{2})^5$$

③ $\iint_S 3y z^5 (y^5+z)\,dy\wedge dz + x z^5 (x^5-z)\,dz\wedge dx + (4+xy(x^5+y^5))\,dx\wedge dy$

P Q R

$S: x^2+3y^2+2z^6$

$2x+y \le 8$

$x \ge 0$

$3 \le y \le 7$

$z \ge 0$

$z^5 = (x^2+3y^2+2)^{5/6}$

$S: \begin{cases} x = x \\ y = y \\ z = \sqrt[6]{x^2+3y^2+2} \end{cases}$

$D(x,y): 3 \le y \le 7,\ 0 \le x \le \frac{8-y}{2}$

$\frac{\partial z}{\partial x} = (x^2+3y^2+2)^{-5/6} \cdot \frac{1}{6} \cdot 2x$

$\frac{\partial z}{\partial y} = (x^2+3y^2+2)^{-5/6} \cdot y$

$N = \left(\frac{x}{3(x^2+3y^2+2)^{5/6}}, \frac{y}{(x^2+3y^2+2)^{5/6}}, 1 \right)$

$\iint_D 3y \cdot z^5 (y^5+z) \cdot \left(-\frac{x}{3(x^2+3y^2+2)^{5/6}} \right) +$

$+ x z^5 (x^5-z) \cdot \left(-\frac{y}{(x^2+3y^2+2)^{5/6}} \right) +$

$+ 4 + xy(x^5+y^5) \cdot 2$

$$= -x4(4^5+2) - x4(x^5-2) + 4 + x4(x^5+4^5) =$$

$$= -x4(x^5+4^5) + 4 + x4(x^5+4^5) =$$

$$= 4$$

$$\Rightarrow S = \iint_D 4\,dx\,dy =$$

$$= 4\left(\int_3^7 \int_0^{8-4/2} dx\right) dy =$$

$$= 4 \cdot \int_3^7 \frac{8-4}{2}\,dy =$$

$$= 2\left(84 - \frac{4}{2}\Big|_3^7\right) =$$

$$= 2\left(56 - \frac{49}{2} - 24 + \frac{9}{2}\right) =$$

$$= 2\left(32 - 20\right) =$$

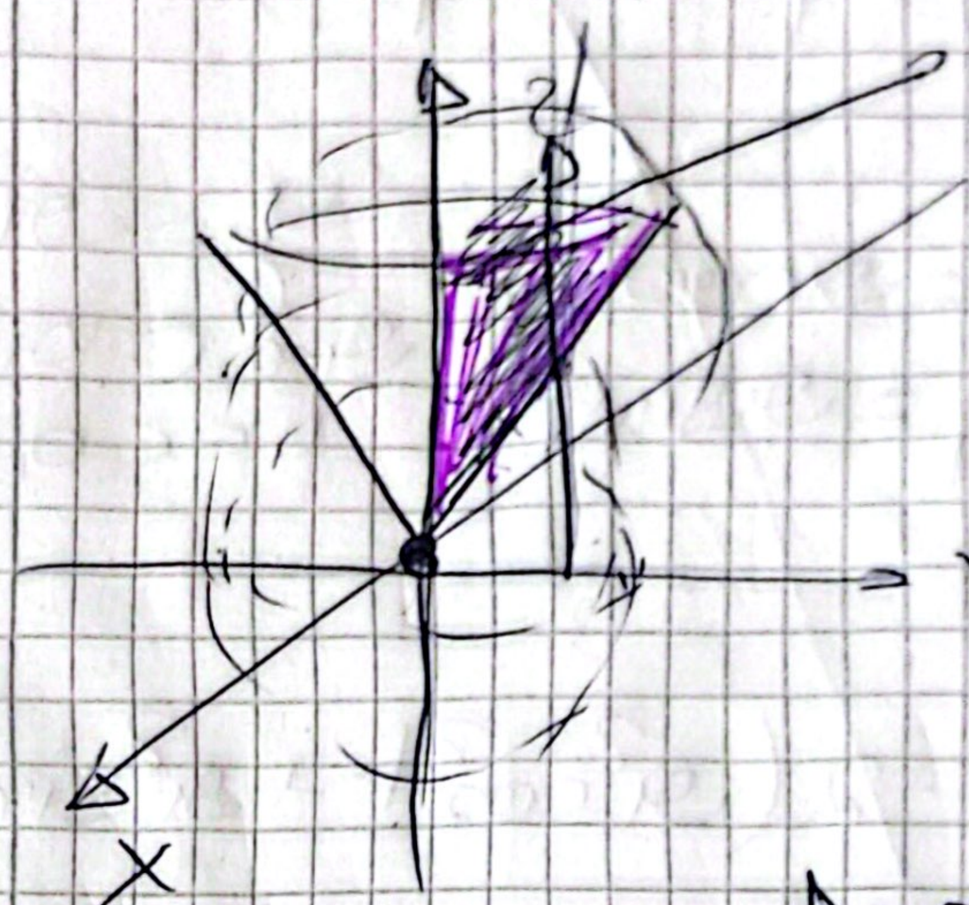
$$= 2 \cdot 12 = 24$$

Sfera ext:

① $K=\{x^2+y^2+z^2\leq 8$

$x^2+y^2\leq z^2$

$z\geq 0\}$



$x^2+y^2=z^2$

$2z^2=8$

$z=\pm 2$, $z>0$ $\Rightarrow z=2$

$\Delta=pr_{xOy}=\{x^2+y^2\leq 4\}$

$P=x-y^2$

$Q=y-z^2$

$R=z-x^2$

$\begin{cases} x=\rho\cos t \\ y=\rho\sin t \\ z=\rho \end{cases}$

$\rho\in[0,2]$

$t\in[0,2\pi]$

$\operatorname{div}\vec{V}=\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}=$

$=1+1+1=3$

$Flux=\iiint_\Delta 3\,dx\,dy\,dz=$

$=6\int_0^{2\pi}\left(\int_0^2 \rho\,d\rho\right)dt=$

$=6\int_0^{2\pi}\frac{\rho^2}{2}\Big|_0^2 dt=$

$=6\int_0^{2\pi}2\,dt=$

$=6\cdot 2\cdot 2\pi=24\pi$

$$\text{Flux} = \iiint_{\sqrt{x^2+y^2}}^{\sqrt{8-x^2-y^2}} 3 \, dz\,dx\,dy =$$

$$= 3 \iint_{\Delta} \sqrt{8-x^2-y^2} - \sqrt{x^2+y^2} \, dx\,dy =$$

$$= \begin{cases} x = r\cos t \\ y = r\sin t \\ y = r \end{cases}$$

$$r \in [0,2]$$
$$t \in [0,2\pi]$$

$$= 3 \int_0^2 \int_0^{2\pi} \left(\sqrt{8-r^2} - \sqrt{r^2} \right) r \, dt\,dr =$$

$$= 2\pi \int_0^2 r\sqrt{8-r^2} - r^2 \, dr =$$

$$= 2\pi \int_0^2 r\sqrt{8-r^2}\,dr - \int_0^2 r^2\,dr =$$

$$\int_0^2 r\sqrt{8-r^2}\,dr = -\frac{1}{2}\int_8^4 \sqrt{u}\,du =$$

$$8 - r^2 = u$$
$$-2r\,dr = du$$

$$= +\frac{1}{2}\int_4^8 u^{1/2}\,du =$$

$$= \frac{1}{2}\cdot\frac{2}{3}u^{3/2}\Big|_4^8 =$$

$$= \frac{1}{3}\left(8^{3/2} - 4^{3/2}\right) =$$

$$= \frac{1}{3}8^{3/2} - 4^{3/2}$$

$$= 2\pi \cdot \left[\frac{1}{3}\left(8^{3/2} - 4^{3/2}\right) - \frac{8}{3} \right] =$$

⑤ $\iiint_V z^2\sqrt{(x^2+y^2+z^2)^3}\,dx\,dy\,dz$

$$\begin{cases} x = \rho\sin\theta\cos\varphi \\ y = \rho\sin\theta\sin\varphi \\ z = \rho\cos\theta \end{cases} \qquad y = \rho^2\sin^2\theta$$

$$D = \{x^2+y^2+z^2 \le \sqrt{3}^2\} \qquad \begin{cases} \rho \in [0,\sqrt{3}] \\ \varphi \in [0,2\pi] \\ \theta \in [0,2\pi] \end{cases}$$

$$\int_0^{2\pi}\int_0^{2\pi}\int_0^{\sqrt{3}} \rho^2\cos^2\theta\sqrt{(\rho^2\sin^2\theta(\cos^2\varphi+\sin^2\varphi)+\rho^2\cos^2\theta)^3}\cdot\rho^2\sin\theta\,d\rho\,d\varphi\,d\theta =$$

$$= \int_0^{2\pi}\int_0^{2\pi}\int_0^{\sqrt{3}} \rho^2\cos^2\theta\sqrt{(\rho^2)^3}\cdot\rho^2\sin\theta\,d\rho\,d\varphi\,d\theta =$$

$$= \int_0^{2\pi}\int_0^{2\pi}\int_0^{\sqrt{3}} \rho^7\cos^2\theta\cdot\sin\theta\,d\rho\,d\varphi\,d\theta =$$

$$= \int_0^{2\pi}\int_0^{2\pi} \frac{81}{8}\cos^2\theta\sin\theta\,d\theta\,d\varphi =$$

$$= 2\pi\cdot\frac{81}{8}\int_{-1}^{1} u^2\,du = 2\pi\cdot\frac{81}{8}\cdot\frac{2}{3} = \frac{81}{12}\pi$$

$$\cos\theta = u$$

$$du = -\sin\theta\,d\theta$$

① $\int_0^1 x^3 \sqrt{e^{x^3}}$

$$R^n(x) = \frac{f^{(n+1)}(0)}{(n+1)!} x^{n+1}$$

$$f(x) = f(0) + f'(0)x + \dots + \frac{f^{(n)}(0)}{n!}x^n + R^n(x)$$

$$e^x = 1 + x + \frac{x^2}{2} + \dots$$

$$e^{\frac{x^3}{2}} = 1 + \frac{x^3}{2} + \frac{(\frac{x^3}{2})^2}{2!} + \dots + \frac{(\frac{x^3}{2})^n}{n!} + R_n(x)$$

$$R_n(x) = \frac{e^c}{(n+1)!}\left(\frac{x^3}{2}\right)^{n+1} \qquad c \in (0, \frac{x^3}{2})$$

$$x^3 e^{\frac{x^3}{2}} = x^3\left(1 + \frac{x^3}{2} + \dots + \frac{(\frac{x^3}{2})^n}{n!}\right) + x^3 R_n(x)$$

$$x^3 R_n(x) = \frac{e^c x^{3(n+2)}}{2^{n+1}(n+1)!} \qquad c \in (0, \frac{1}{2})$$

$$e^c \le e^{1/2} < 2$$

$$x^3 R_n(x) \le \frac{2x^{3(n+2)}}{2^{n+1}(n+1)!} = \frac{x^{3(n+2)}}{2^n(n+1)!}$$

$$\int_0^1 x^3 R_n(x)\,dx \le \int_0^1 \frac{x^{3(n+2)}}{2^n(n+1)!} =$$

$$= \frac{1}{2^n(n+1)!} \cdot \frac{1}{3(n+2)+1}$$

$$= \frac{1}{2^n(n+1)!(3n+7)}$$

$n=0 \Rightarrow \alpha = \frac{1}{7}$

$n=1 \Rightarrow \alpha_2 \frac{1}{2 \cdot 2! \cdot 10} = \frac{1}{40}$

$n=2 \Rightarrow \alpha = \frac{1}{4 \cdot 3! \cdot 13} = \frac{1}{312} < \frac{1}{300}$

$P_2(x) = 1 + \frac{x^3}{2} + \frac{(x^3+2)^2}{2!} = \ldots + \frac{x^3}{2} + \frac{x^6}{8} \Big| \cdot x^3$

$x^3 \cdot P_2(x) = x^3 + \frac{x^6}{2} + \frac{x^9}{8} \Big|_0^1 =$

$= \frac{1}{4} + \frac{1}{14} + \frac{1}{80} \approx 0{,}334$

$0 < I - \sum_{k=0}^{2} \frac{1}{(3k+4) \cdot 2^k \cdot k!} \leq \alpha_2$

$\sum_{k=0}^{2} \frac{1}{(3k+4) 2^k k!} \leq I \leq \alpha_2 + \sum$

$\frac{1}{4} + \frac{1}{14} + \frac{1}{80} \leq \frac{1}{4} + \frac{1}{14} + \frac{1}{80} + \frac{1}{312}$

$\frac{187}{560} \qquad \frac{187}{560}$

$0{,}339 \leq I \leq 0{,}3371$

① Cum Leibniz:

$$\sum_{n\geq 3} \frac{(n-1)^2}{n^p} \cdot \sqrt[3]{\frac{n}{n}}$$

$$\lim_{n\to\infty} n^\alpha \frac{(n^2+2n+1)}{n^p} \cdot \sqrt[3]{\frac{1}{n}} \cdot \left(\frac{n}{n}\right)^{\frac{1}{3}}$$

$$\alpha + 2 - p \Rightarrow \alpha + \frac{3}{2} - \frac{1}{3} > p$$

$$\alpha = p - \frac{5}{3} < 1 \quad p < \frac{8}{3} \quad SA$$

$$p > \frac{8}{3} \quad SC.$$

② Pt. orice alte lui x, seria de puteri e conv.:

$$\sum_{n\geq 1} \left(\frac{n}{1+7n}\right)^n x^n$$

$$a_n = \left(\frac{n}{1+7n}\right)^n$$

$$L = \lim_{n\to\infty} \sqrt{a_n} = \lim_{n\to\infty} \frac{n}{1+7n} = \frac{1}{7}$$

$$\Rightarrow R = \frac{1}{L} = 7$$

pt. $x = 7 \Rightarrow \sum \left(\frac{n}{1+7n}\right)^n \cdot \underbrace{7^n}_{x^n}$

$$\lim_{n\to\infty} \frac{x^{n+1}}{x^n} = \lim_{n\to\infty} \frac{n^{n+1} \cdot 7^{n+1}}{(1+7n)^{n+1}} \cdot \frac{(1+7n)^n}{7^n \cdot n^n} =$$

$$\to \lim_{n\to\infty} \frac{7n}{1+7n} = 1$$

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③ Bài 2 đề cô Nga nhé, mọi người cố lên

$A = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 \le 4x+2y-19\}$ +.

$f(x,y) = x^2+y^2+2x+15$ $\left|\begin{array}{l}\frac{\partial f}{\partial x} = -1 \\ \frac{\partial f}{\partial y} = 0\end{array}\right| \Rightarrow (-1,$

$\frac{\partial f}{\partial x}$

$L(x,y,\lambda) = f(x,y) + \lambda g(x,y) =$ Lagrange

$= x^2+y^2+2x+15+\lambda(x^2+y^2-4x-2y+19)$

$\begin{cases} \frac{\partial L}{\partial x} = 2x+2+2x\lambda-4\lambda = 0 \quad (1) \\ \frac{\partial L}{\partial y} = 2y+2y\lambda-8\lambda = 0 \quad (2) \\ \frac{\partial L}{\partial \lambda} = 0 \Rightarrow x^2+4x+y^2-8y+19 = 0 \end{cases}$

(1) ~~$2(x+2)$~~

(2) $y+y\lambda-4\lambda-4\lambda = 0$

$y(1+\lambda) = 4\lambda \Rightarrow \boxed{y = \frac{4\lambda}{1+\lambda}}$

(1) $x+1+x\lambda-2\lambda = 0$

$x(1+\lambda) = 2\lambda-1$

$\boxed{x = \frac{2\lambda-1}{1+\lambda}}$

$\left(\frac{2\lambda-1}{1+\lambda}\right)^2 - 4\cdot\frac{2\lambda-1}{1+\lambda} + \frac{16\lambda^2}{(1+\lambda)^2} - 8\cdot\frac{4\lambda}{1+\lambda} + 19 = 0$

$\frac{(1+\lambda)^2}{(1+\lambda)^2}$ $\frac{1+\lambda}{1+\lambda}$

$4\lambda^2 - 4\lambda + 1 = (1+2\lambda+\lambda^2)(2\lambda-1) + 16\lambda^2$

$-32\lambda(1+\lambda)$

$$4R^2 - 4R + 1 - 4(2R-1)(R+1) + 16R^2 - 32R(1+R) + 19(1+2R+R^2) = 0$$

$$4R^2 - 4R + 1 - 4(2R^2 + R - 1) + 16R^2 - 32R^2 + 19R^2 + 38R + 19 = 0$$

$$4R^2 + 2R + 20 - 8R^2 - 4R + 4 = 0$$

$$-R^2 - 2R + 24 = 0$$

$$R^2 + 2R - 24 = 0$$

$$\Delta = 4 + 4 \cdot 24 = 100$$

$$R_{1,2} = \frac{-2 \pm 10}{2} \begin{cases} -6 \\ 4 \end{cases}$$

$$p_t \cdot R_1 = 4 \Rightarrow y = \frac{4 \cdot 4}{5} = \frac{16}{5}$$

$$x = \frac{2 \cdot 4 - 1}{5} = \frac{7}{5}$$

$$A\left(\frac{7}{5}, \frac{16}{5}\right) = \frac{49}{25} + \frac{256}{25} + \frac{14}{5} + \frac{25}{5} =$$

$$= 30 \Rightarrow \text{MINIM}$$

$$p_t \cdot R_2 = -6 \Rightarrow x = \frac{-13}{-5} = \frac{13}{5}$$

$$y = \frac{-24}{-5} = \frac{24}{5}$$

$$A\left(\frac{13}{5}, \frac{24}{5}\right) = \frac{169}{25} + \frac{576}{25} + \frac{26}{5} + 15 = 50$$

MAX.

PV De ext.

③ $f(x,y,z) = x^3 + 5y^3 - 3x^2y + 4z^2 - 12y - 8$

$$\frac{\partial f}{\partial x} = 3x^2 + 6xy = 0 \Rightarrow \begin{cases} x = 2y \\ x = 0 \end{cases}$$

$$\frac{\partial f}{\partial y} = 15y^2 - 3x^2 + z^2 - 12$$

$$\frac{\partial f}{\partial z} = 2yz - 6 = 0 \Rightarrow z = \frac{3}{y}$$

I $x = 0 \Rightarrow 15y^2 + z^2 - 12 = 0$

$$15y - 12y^2 + 9 = 0$$

$y \neq 0$

II $x = 2y \Rightarrow 15y^2 + 3(2y)^2 + z^2 - 12 =$

$$= 3y^2 + \frac{9}{y^2} - 12 = 0$$

$$\Rightarrow 3y^4 - 12y^2 + 9 = 0$$

$t = y^2$

$$\Rightarrow t^2 - 4t + 3 = 0 \begin{cases} t = 1 \Rightarrow y = \pm 1 \\ t = 3 \Rightarrow y = \pm\sqrt{3} \end{cases}$$

$$\Rightarrow \begin{cases} (2, 1, 3) \\ (-2, -1, -3) \\ (2\sqrt{3}, \sqrt{3}, \sqrt{3}) \\ (-2\sqrt{3}, -\sqrt{3}, -\sqrt{3}) \end{cases} \Rightarrow \text{Hessiana}$$

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(ε) $-\lambda_0 \frac{\partial^2 \Omega}{\partial z^2}\Big|_0^2 = -\rho_0 \bar{u}$.

$$\iint_{x^2+y^2\le 4} -\frac{\mu}{6}\frac{\partial^2 u}{\partial z^2}\,dx\,dy = \frac{\mu}{6}\cdot\frac{\rho_0 g}{\mu}\cdot\pi\cdot 2^2 = \frac{N\pi}{\left(\frac{1}{\mu}\right)\left(\frac{\rho_0}{\mu}\right)}$$

$$\frac{\partial P}{\partial x} = +\frac{1}{h} \qquad \frac{\partial P}{\partial y} = +\frac{\rho g}{h}$$

$$-\frac{\partial P}{\partial x} = \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\vec{j} + \left(\frac{\partial Q}{\partial z} - \frac{\partial R}{\partial y}\right)\vec{k}$$

$$= \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial z}\right)\vec{i} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\vec{j} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\vec{k} =$$

$$= (-b-2)\vec{i} + (-a+2)\vec{j} + (-a-b)\vec{k} =$$

$$\operatorname{curl}\vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} =$$

$$\begin{cases} x+2y+z=12 \\ x^2+y^2=4^2 \end{cases} \Rightarrow z = 12-x-2y$$

$$\begin{cases} x=x \\ y=y \end{cases}$$

$$\textcircled{4} \iint_S (xb+2y-bz)\,dx + \tfrac{1}{2}(-bx+y^2+2z)\,dy$$

$$+ (2x-2y+2b)\,dz$$

⑤ $\iint_S (x-2)^3 + y^3 + z^3 \, dy \wedge dz + dz \wedge dx + dx \wedge dy$

S

$S: (x-2)^2 + y^2 + z^2 = 5 \rightarrow (2,0,0)$

$a = \sqrt{5}$

$P = Q = R = (x-2)^3 + y^3 + z^3$

$\frac{\partial P}{\partial x} = 3(x-2)^2$

$\frac{\partial Q}{\partial y} = 3y^2$

$\frac{\partial R}{\partial z} = 3z^2$

$\iiint 3(x-2)^2 + \underbrace{3y^2 + 3z^2}_{3 \cdot a^2} \, dx \, dy \, dz =$

$x = 2 + \rho$

$y = \rho \sin\theta \cos\varphi$

$z = \rho \cos\theta$

$x - 2 = \rho \sin\theta \cos\varphi$

$y = \rho \sin\theta \cos\varphi$

$z = \rho \cos\theta$

$y = \rho^2 \sin\theta$

$\rho \in [0, \sqrt{5}]$

$\theta \in [0, \pi]$

$\varphi \in [0, 2\pi]$

$3 \int_0^{2\pi} \int_0^{\pi} \int_0^{\sqrt{5}} \rho^2 (\rho^2 \sin\theta) \, d\rho \, d\theta \, d\varphi =$

$= 3 \cdot 2\pi \cdot \int_0^{\pi} \frac{\rho^5}{5} \Big|_0^{\sqrt{5}} \sin\theta =$

$= 6\pi \cdot \frac{(\sqrt{5})^5}{5} (-\cos\theta) \Big|_0^{\pi} =$

$= 60\pi\sqrt{5}$

8ub 2023

① $a_n = \left(\frac{1}{\sqrt{n+2}+\sqrt{n+1}}\right)^{\sin\frac{1}{n}}$

$$\lim_{n\to\infty} \frac{\sqrt{n+2}-\sqrt{n+1}}{n+2-n-1} \cdot \frac{\sin\frac{1}{n}}{\frac{1}{n}} \cdot \frac{1}{n^{\alpha}} =$$

$$= \lim_{n\to\infty} \frac{n^{1/2}\left(\sqrt{1+\frac{2}{n}}-\sqrt{1+\frac{1}{n}}\right)}{1} \cdot \frac{1}{n^{\alpha}}$$

$$\Rightarrow \alpha + \frac{1}{2} = 1$$

$$\alpha = \frac{1}{2} < 1 \Rightarrow \text{[illegible]} \quad \underset{\sim}{\text{Div}}$$

②

$$f(x,y) = x^2+2xy-4x-4$$

$$0 \le x \le 2, \quad -1 \le y \le 3-2x$$

Pct. Int D

$$\frac{\partial f}{\partial x} = 2x+2y-4=0$$

$$\frac{\partial f}{\partial y} = 2x-1=0 \Rightarrow x = \frac{1}{2}$$

$y = \frac{4-1}{2} = \frac{3}{2}$

$\in [0,2]$

$$-1 \le \frac{3}{2} \le 2 \quad \underset{\sim}{\text{Adev}}$$

$$f\left(\frac{1}{2}, \frac{3}{2}\right) = \frac{1}{4} + 2\cdot\frac{1}{2}\cdot\frac{3}{2} - 4\cdot\frac{1}{2} - \frac{3}{2} = \in \text{Int D}$$

$$= \frac{-3}{2} = \frac{7}{4} - \frac{4}{2} = -\frac{9}{4}$$

Pe Fr. Lui

$x=0 \Rightarrow y \in [-1,3]$

$\Downarrow$

$y=3 \to -3$

$y=1 \to -1$

$y=-1 \Rightarrow x \in [0,2] \longrightarrow x^2-6x+1$

$x=0 \to 1$

$x=2 \to -7$

$y=3-2x \Rightarrow x \in [0,2]$

$f(x,3-2x) = -3x^2+4x-3$

$x=0 \to -3$

$x=2 \to -7$

③

$x^2+y^2=1-z \to \min \min x^2+y^2=1$

$z=1-x^2-y^2$

$\frac{\partial z}{\partial x} = -2x \qquad \frac{\partial z}{\partial y} = -2y$

$D = pr_{xOy} \begin{cases} x^2+y^2 \le 1 \\ y \in [0,1] \end{cases}$

$x = y\cos t$

$y = y\sin t$

$y = y$

$A = \iint \sqrt{1+\left(\frac{\partial z}{\partial x}\right)^2+\left(\frac{\partial z}{\partial y}\right)^2} =$

$= \iint \sqrt{1+4x^2+4y^2} = \iint \sqrt{1+4(x^2+y^2)}$

$$\Rightarrow \int_0^{2\pi}\int_0^{1} r\sqrt{1+4r^2}\,dr\,d\theta \quad (*)$$

$$1+4r^2=t \qquad r=0 \Rightarrow t=1$$

$$8r\,dr=dt \qquad r=1 \Rightarrow t=5$$

$$4r\,dr=dt$$

$$\frac{dt}{8}$$

(2)

$$\int_0^{2\pi}\int_1^{5} \sqrt{t}\,\frac{dt}{8}$$

$$= \int_0^{2\pi}\int_1^{5} t^{1/2}\,\frac{dt}{8}$$

$$= \int_0^{2\pi} \frac{1}{8}\cdot\frac{t^{3/2}}{3/2}\Big|_1^5 \,d\theta$$

$$= 2\pi\cdot\frac{1}{8}\left(\frac{5^{3/2}\cdot 2}{3}-\frac{1\cdot 2}{3}\right)$$

$$= \frac{\pi}{6}\left(5^{3/2}-1\right)$$

(4)

$$x^2+y^2+z=1$$

$$x^2+y^2=1$$

$$z=1$$

$$\Rightarrow \text{volum} ?$$

$$x^2+y^2\le 1$$

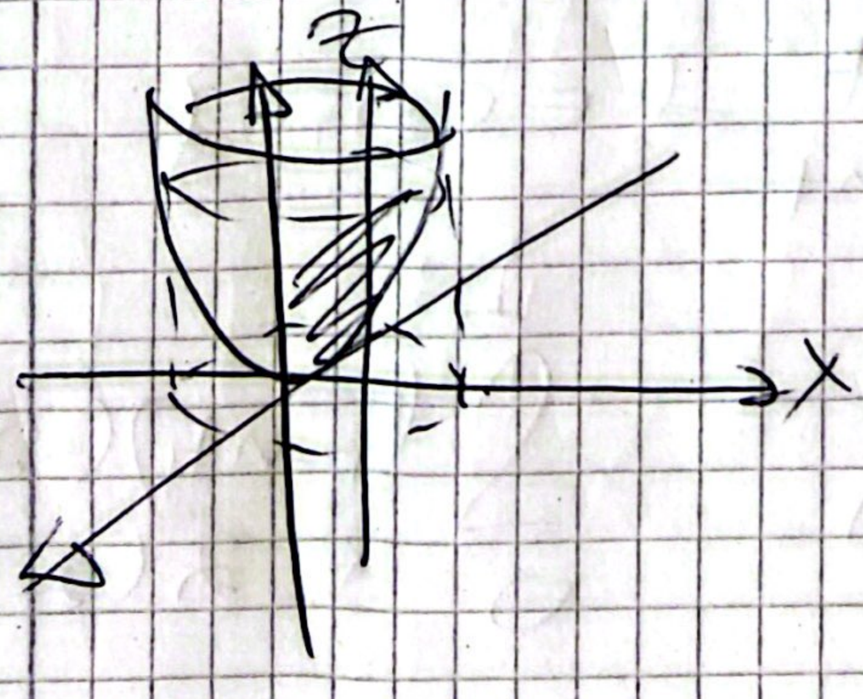
$$x=r\cos\theta$$

$$y=r\sin\theta$$

$$y=\theta$$

$$\int_{-1}^{1-x^2-y^2}\iint dz\,dx\,dy =$$

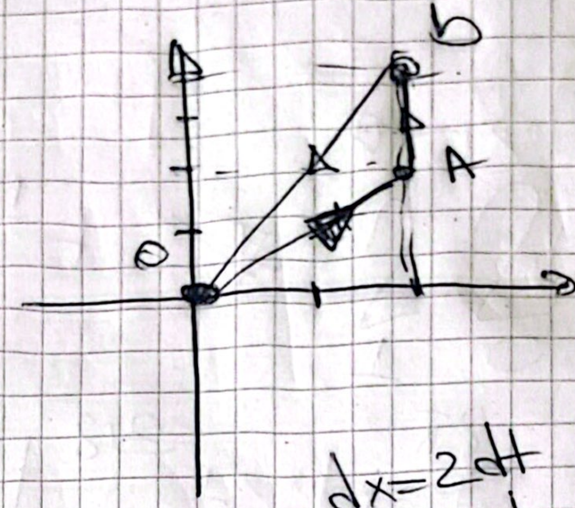
$$= \iint (1-x^2-y^2)+1 \quad (*)$$



④ $2\pi \int_0^1 (2-x^2)\,dx > \frac{3\pi}{2}$

⑤ $\vec{F} = \overset{P}{xy}\vec{i} + \overset{Q}{2x^2y}\vec{j}$ $\int$ $\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$

O(0,0)
A(2,2)
B(2,4)



$O \to A$

$x(t) = (1-t)\cdot 0 + t\cdot 2 = 2t \Rightarrow dx = 2dt$
$y(t) = (1-t)0 + 2t = 2t \Rightarrow dy = 2dt$
$t \in [0,1]$

$\int_0^1 16(t^3+t^2)\,dt = \frac{28}{3}$

$A \to B$

$x(t) = (1-t)\cdot 2 + 2t = 2 \Rightarrow dx = 0$
$y(t) = (1-t)\cdot 2 + 4t = 2t+2 \Rightarrow dy = 2dt$
$t \in [2,4]$

$\int_2^4 2\cdot 2^2\,dy = 16$

$B \to O$

$x(t) = 2-2t \quad t \in [0,1]$
$y(t) = 4-4t$
$dx = -2$
$dy = -4$

$\int_0^1 (1-4t)(-2) + 2(2-2t)(-4)\,dt$

$$t=\sin^2 x \Rightarrow dt=2\sin\cos x\,dx$$

$$I=\int_0^{\frac{\pi}{2}} \sin^{(p-2)}x\cos^{(q-2)}x\,dx$$

$$I=\frac{1}{2}\int_0^1 t^{\frac{(p-2)}{2}}(1-t)^{\frac{(q-2)}{2}}dt$$

$$I=\frac{1}{2}\beta\left(\frac{p}{2},\frac{q}{2}\right)$$

$$I=\frac{1}{2}\frac{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)}{\Gamma\left(\frac{p}{2}+\frac{q}{2}\right)}$$

$$I=\frac{1}{2}\frac{\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{q}{2}\right)}{\Gamma\left(\frac{p+q}{2}\right)}$$

$$\int_0^{\frac{\pi}{2}} \sin^p x\cos^q x\,dx = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^2 x\cos^3 x}{(\sin x+\cos x)^{7/2}}\,dx = \frac{1}{2}\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{4}{2}\right)}{\Gamma\left(\frac{7}{2}\right)}$$

$$= \frac{1}{2}\frac{\frac{1}{2}\sqrt{\pi}\cdot 1}{\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}} = \frac{1}{15}$$

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}x\cos^{-2}x\,dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{1}{2}}x}{\cos^2 x}\,dx$$

$$\int_0^{\frac{\pi}{2}} \tan^{\frac{1}{2}}x\sec^2 x\,dx$$

$$t=\tan x \Rightarrow dt=\sec^2 x\,dx$$

$$\int_0^{\infty} \frac{t^{\frac{1}{2}}}{1+t^2}\,dt$$

$$\int_0^{\infty} \frac{t^{\frac{1}{2}}}{(1+t)^2}\,dt$$

$$\int_0^{\infty} \frac{t^{\frac{3}{2}-1}}{(1+t)^{\frac{3}{2}+\frac{1}{2}}}\,dt$$

$$\int_0^{\infty} \frac{t^{p-1}}{(1+t)^{p+q}}\,dt = \beta(p,q)$$

$$\int_0^{\infty} \frac{t^{\frac{3}{2}-1}}{(1+t)^{\frac{3}{2}+\frac{1}{2}}}\,dt = \beta\left(\frac{3}{2},\frac{1}{2}\right)$$

$$= \frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(2)} = \frac{\frac{1}{2}\sqrt{\pi}\cdot\sqrt{\pi}}{1} = \frac{\pi}{2}$$

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}x\cos^{-\frac{1}{2}}x\,dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{1}{2}}x}{\cos^{\frac{1}{2}}x}\,dx$$

$$\int_0^{\frac{\pi}{2}} \tan^{\frac{1}{2}}x\,dx$$

$$t=\tan x \Rightarrow dt=\sec^2 x\,dx$$

$$\int_0^{\infty} \frac{t^{\frac{1}{2}}}{1+t^2}\,dt$$

$$\int_0^{\infty} \frac{t^{\frac{3}{2}-1}}{(1+t^2)^{\frac{3}{4}+\frac{1}{4}}}\,dt$$

$$\int_0^{\infty} \frac{t^{p-1}}{(1+t^2)^{p+q}}\,dt = \frac{1}{2}\beta\left(\frac{p}{2},\frac{q}{2}\right)$$

$$\int_0^{\infty} \frac{t^{\frac{3}{2}-1}}{(1+t^2)^{\frac{3}{4}+\frac{1}{4}}}\,dt = \frac{1}{2}\beta\left(\frac{3}{4},\frac{1}{4}\right)$$

$$= \frac{1}{2}\frac{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{1}{4}\right)}{\Gamma(1)} = \frac{1}{2}\Gamma\left(\frac{3}{4}\right)\Gamma\left(1-\frac{3}{4}\right)$$

$$= \frac{1}{2}\frac{\pi}{\sin\frac{\pi}{4}} = \frac{\pi}{\sqrt{2}}$$

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}x\cos^{\frac{1}{2}}x\,dx$$

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}x\sqrt{1+\tan^2 x}\,dx$$

$$\int_0^{\frac{\pi}{2}} \sin^{\frac{1}{2}}x\sec x\,dx$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{1}{2}}x}{\cos x}\,dx$$

$$\int_0^{\frac{\pi}{2}} \frac{2\sin^{\frac{1}{2}}x\cos x}{2\cos^2 x}\,dx$$

$$t=\sin x \Rightarrow dt=\cos x\,dx$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{2-t^2}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{1+1-t^2}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{1+(1-t^2)}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{1+(1-t)(1+t)}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{1+(1-t)(1+t)}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{(1-t)+(1-t)(1+t)}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{(1-t)[1+(1+t)]}\,dt$$

$$\int_0^1 \frac{2t^{\frac{1}{2}}}{(1-t)(2+t)}\,dt$$

$$\int_0^1 \frac{t^{\frac{1}{2}}}{(1-t)(2+t)}\,dt$$

$$\int_0^1 \frac{t^{\frac{3}{2}-1}}{(1-t)^{\frac{1}{2}}(2+t)}\,dt$$

$$\int_0^1 \frac{t^{\frac{3}{2}-1}}{(1-t)^{\frac{1}{2}}(2+t)^{\frac{1}{2}+\frac{1}{2}}}\,dt$$

$$\int_0^1 \frac{t^{p-1}}{(1-t)^{q}(2+t)^{p+q}}\,dt = \frac{1}{2}\beta\left(\frac{p}{2},\frac{q}{2}\right)$$

$$\int_0^1 \frac{t^{\frac{3}{2}-1}}{(1-t)^{\frac{1}{2}}(2+t)^{\frac{1}{2}+\frac{1}{2}}}\,dt = \frac{1}{2}\beta\left(\frac{3}{2},\frac{1}{2}\right)$$

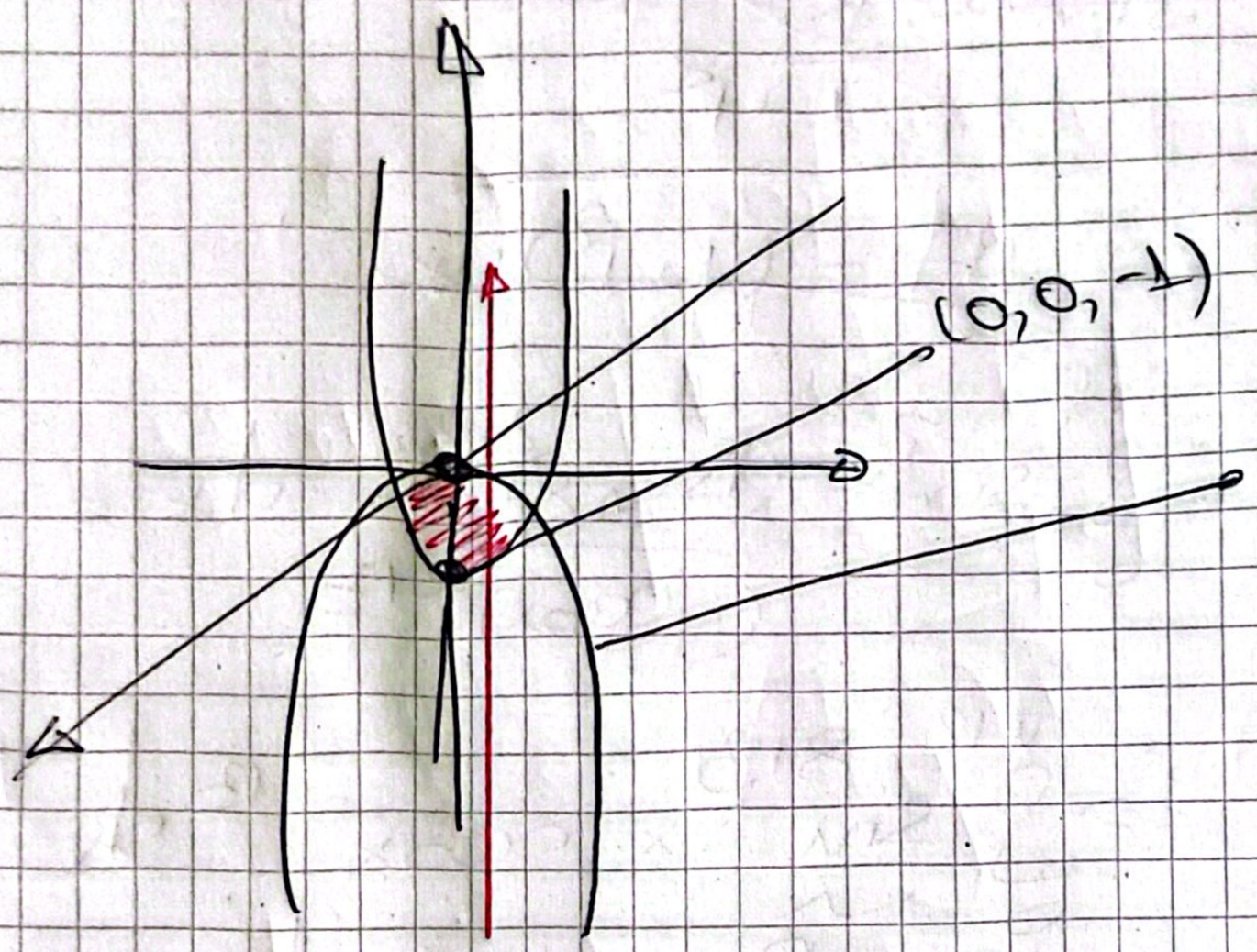
$$= \frac{1}{2}\frac{\Gamma\left(\frac{3}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma(2)} = \frac{1}{2}\frac{\frac{1}{2}\sqrt{\pi}\cdot\sqrt{\pi}}{1} = \frac{\pi}{4}$$

② Volumul mărginit de

$S_1: z = -x^2 - y^2 \Rightarrow x^2 + y^2 + z = 0$

$S_2: z = x^2 + y^2 - 1 \Rightarrow x^2 + y^2 = (z+1)$

con $(0,0,1)$



$x^2 + y^2 - 1 = -x^2 - y^2$

$2x^2 + 2y^2 = 1$

$x^2 + y^2 = \frac{1}{2} \rightarrow D = proy_{xOy}$

$z + 1 = -z$

$2z = -1$

$z = -\frac{1}{2}$

$$\iiint_{-x^2-y^2}^{-\frac{1}{2}} dz\,dx\,dy = \iint_{x^2+y^2-1} x^2 + y^2 - 1 + x^2 + y^2\,dx\,dy =$$

$$= \iint 2(x^2+y^2) - 1$$

(2) $x = r\cos t$
$y = r\sin t$

$r \in [0, \frac{1}{2}]$
$t \in [0, 2\pi]$

$$\int_0^{2\pi} \int_0^{\frac{1}{2}} (2r^2 - 1)\, r\, dr\, dt =$$

$$= \frac{\pi}{4}$$

(3) $x^2 + y^2 + z^2 = 1 \Rightarrow z = \sqrt{1 - x^2 - y^2}$

$x^2 + y^2 = x \Rightarrow \left(x - \frac{1}{2}\right)^2 + y^2 - \frac{1}{4} = 0$

$\left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$

$x = \frac{1}{2}\cos t$
$y = \frac{1}{2}\sin t$

$x^2 + y^2 \le x$
$r^2 \le r\cos t$
$r \ge 0$
$\cos t \in [0, \frac{\pi}{2}]$

$$\frac{\partial z}{\partial x} = -\frac{2x}{2\sqrt{1 - x^2 - y^2}} \qquad \frac{\partial z}{\partial y} = -\frac{y}{\sqrt{1 - x^2 - y^2}}$$

$$A = \iint \sqrt{1 + \frac{x^2 + y^2}{1 - x^2 - y^2}}\, dx\, dy =$$

$$= \iint \sqrt{\frac{1 - x^2 - y^2 + x^2 + y^2}{1 - x^2 - y^2}}\, dx\, dy =$$

$$= 2\pi \cdot \int_0^{\frac{\pi}{2}\cos t} \cdots$$

$$\frac{\partial z}{\partial x} = \frac{-1}{2\sqrt{1-x^2-y^2}}(-2x) = \frac{x}{\sqrt{1-x^2-y^2}}$$

$$\frac{\partial z}{\partial y} = \frac{-1}{2\sqrt{1-x^2-y^2}}(-2y) = \frac{y}{\sqrt{1-x^2-y^2}}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\sqrt{1-x^2-y^2}(1) - x\left(\frac{1}{2\sqrt{1-x^2-y^2}}(-2x)\right)}{1-x^2-y^2}$$

$$= \frac{\sqrt{1-x^2-y^2} + \frac{x^2}{\sqrt{1-x^2-y^2}}}{1-x^2-y^2} = \frac{1-x^2-y^2+x^2}{(1-x^2-y^2)^{3/2}} = \frac{1-y^2}{(1-x^2-y^2)^{3/2}}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\sqrt{1-x^2-y^2}(1) - y\left(\frac{1}{2\sqrt{1-x^2-y^2}}(-2y)\right)}{1-x^2-y^2}$$

$$= \frac{\sqrt{1-x^2-y^2} + \frac{y^2}{\sqrt{1-x^2-y^2}}}{1-x^2-y^2} = \frac{1-x^2-y^2+y^2}{(1-x^2-y^2)^{3/2}} = \frac{1-x^2}{(1-x^2-y^2)^{3/2}}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-1}{2}(1-x^2-y^2)^{-3/2}(-2y)(-2x) = \frac{-xy}{(1-x^2-y^2)^{3/2}}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{-1}{2}(1-x^2-y^2)^{-3/2}(-2x)(-2y) = \frac{-xy}{(1-x^2-y^2)^{3/2}}$$

$$\frac{\partial^2 z}{\partial x^2}\frac{\partial^2 z}{\partial y^2} - \left(\frac{\partial^2 z}{\partial x \partial y}\right)^2 = \frac{(1-y^2)(1-x^2) - x^2y^2}{(1-x^2-y^2)^3} = \frac{1-x^2-y^2+x^2y^2-x^2y^2}{(1-x^2-y^2)^3} = \frac{1-x^2-y^2}{(1-x^2-y^2)^3} = \frac{1}{(1-x^2-y^2)^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{1-y^2}{(1-x^2-y^2)^{3/2}} > 0$$

$$\frac{\partial^2 z}{\partial x^2}\frac{\partial^2 z}{\partial y^2} - \left(\frac{\partial^2 z}{\partial x \partial y}\right)^2 > 0$$

$$\frac{\partial z}{\partial x} = 0 \Rightarrow x = 0$$

$$\frac{\partial z}{\partial y} = 0 \Rightarrow y = 0$$

$$(0,0,1)$$

$$f(x,y) = x^2 + 2xy + 3y^2 - 2x - 6y + 4$$

$$\frac{\partial f}{\partial x} = 2x + 2y - 2 = 0 \Rightarrow x + y = 1$$

$$\frac{\partial f}{\partial y} = 2x + 6y - 6 = 0 \Rightarrow x + 3y = 3$$

$$y = 2$$

$$x = -1$$

$$(-1, 2)$$

$$\frac{\partial^2 f}{\partial x^2} = 2$$

$$\frac{\partial^2 f}{\partial y^2} = 6$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2$$

$$\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 = 12 - 4 = 8 > 0$$

$$\frac{\partial^2 f}{\partial x^2} = 2 > 0$$

$$f(-1, 2) = 1 + 4 - 12 + 2 + 12 + 4 = 11$$

$$(-1, 2, 11)$$

$$f(x,y) = x^2 + y^2 + 2xy + 4x - 4y + 1$$

$$\frac{\partial f}{\partial x} = 2x + 2y + 4 = 0 \Rightarrow x + y = -2$$

$$\frac{\partial f}{\partial y} = 2y + 2x - 4 = 0 \Rightarrow y + x = 2$$

$$0 = -4$$

$$\frac{\partial^2 f}{\partial x^2} = 2$$

$$\frac{\partial^2 f}{\partial y^2} = 2$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2$$

$$\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 = 4 - 4 = 0$$

$$f(x,y) = x^3 + y^3 - 3x - 12y + 20$$

$$\frac{\partial f}{\partial x} = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\frac{\partial f}{\partial y} = 3y^2 - 12 = 0 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$$

$$(1, 2), (1, -2), (-1, 2), (-1, -2)$$

$$\frac{\partial^2 f}{\partial x^2} = 6x$$

$$\frac{\partial^2 f}{\partial y^2} = 6y$$

$$\frac{\partial^2 f}{\partial x \partial y} = 0$$

$$\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 = 36xy$$

$$\frac{\partial^2 f}{\partial x^2} = 6x > 0$$

$$f(1, 2) = 1 + 8 - 3 - 24 + 20 = 2$$

$$(1, 2, 2)$$

$$\frac{\partial^2 f}{\partial x^2} = 6x < 0$$

$$f(-1, -2) = -1 - 8 + 3 + 24 + 20 = 38$$

$$(-1, -2, 38)$$

$$\frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 = -36xy$$

$$\frac{\partial^2 f}{\partial x^2} = 6x > 0$$

$$f(1, -2) = 1 - 8 - 3 + 24 + 20 = 34$$

$$(1, -2, 34)$$

$$\frac{\partial^2 f}{\partial x^2} = 6x < 0$$

$$f(-1, 2) = -1 + 8 + 3 - 24 + 20 = 6$$

$$(-1, 2, 6)$$